12. Correlation coefficient is the _____ between the regression coefficients

a) arithmetic mean b) geometric mean

c) harmonic mean d) none of the above

13. If the two lines of regression are perpendicular to each other then r_{XY} =

a) 0 b) 1 c) –1 d) 0.5

14. If the two regression lines are parallel then

a) $r_{XY} = 0$ b) $r_{XY} = +1$ c) $r_{XY} = -1$ d) $r_{XY} = \pm 1$

15. Angle between the two regression lines is

a) $\tan^{-1}\left(\frac{m_1 + m_2}{1 - m_1 m_2}\right)$ b) $\tan^{-1}\left(\frac{m_1 m_2}{1 + m_1 m_2}\right)$

c) $\tan^{-1}\left(\frac{m_1 - m_2}{1 + m_1\ m_2}\right)$ d) none of the above

16. b_{XY} =

a) $r_{XY}\frac{SD(Y)}{SD(X)}$ b) $r_{XY}\frac{SD(X)}{SD(Y)}$

c) $r_{XY}\,SD(X)\ SD(Y)$ d) $\frac{1}{b_{YX}}$

17. Regression equation of X on Y is

a) $Y = a + b_{YX}x + e$ b) $Y = b_{XY}x + a + e$

c) $X = a + b_{XY}y + e$ d) $X = b_{YX}y + a + e$

18. For the regression equation $2\hat{Y} = 0.605x + 351.58$. The regression coefficient of Y on X is

a) $b_{XY} = 0.3025$ *b)* $b_{XY} = 0.605$

c) $b_{YX} = 175.79$ d) $b_{YX} = 351.58$

19. If $b_{XY} = 0.7$ and 'a' = 8 then the regression equation of X on Y is

a) $Y = 8 + 0.7\ X$ b) $X = 8 + 0.7\ Y$

c) $Y = 0.7 + 8\ X$ d) $X = 0.7 + 8\ Y$

20. The regression lines intersect at

a) $(\overline{X}, \overline{Y})$ b) (X, Y) c) (0, 0) d) (1, 1)

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